	GLOBAL ECONOMICS **GRADE: 11TH** **MIDTERM (C1–C7)** **HYPOTHESIS TEST DESIGN AND DECISION** **TEACHER'S NAME: Nicolás López Cuéllar**	**TERM III** **2025–2026**

Learning objective: Design and apply hypothesis tests from real contexts by identifying the parameter, writing hypotheses, selecting the test direction, applying a proportion test and a mean test, interpreting statistical errors, and determining the minimum sample size needed to reject the null hypothesis.	**Evaluated criteria:** C1: Describes the concept of a statistical hypothesis. C2: Identifies the null and alternative hypotheses in a statistical test. C3: Determines whether a test should be one-tailed or two-tailed. C4: Applies a hypothesis test in a population proportion setting. C5: Applies a hypothesis test in a population mean counting setting. C6: Identifies Type I or Type II error and its consequences. C7: Finds the minimum sample size needed to reject H_0.

Critical values used

$$z_{0.95} = 1.645 \qquad z_{0.975} = 1.960 \qquad z_{0.99} = 2.326$$

For right-tailed tests, reject when $z > z_{\text{critical}}$. For left-tailed tests, reject when $z < -z_{\text{critical}}$. For two-tailed tests, reject when $|z| > z_{\text{critical}}$.

Formula tools

Mean-count test — Proportion-test

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \qquad z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Minimum sample-size tools:

$$n = \left(\frac{z\sigma}{\bar{x} - \mu_0}\right)^2 \qquad n = \frac{z^2 p_0(1-p_0)}{(\hat{p} - p_0)^2}$$

Task outline. Create one consistent real-world counting scenario that compares two groups, such as Prom 2025 vs Prom 2026, Group A vs Group B, before vs after, region 1 vs region 2, two companies, two cohorts, or two markets. The counted variable must be numerical, such as late arrivals, completed assignments, absences, deliveries, or delays.
The same scenario, groups, counted variable, and binary threshold must be used throughout C1–C7.

Required data table format. Show how the data table is organized so that the counting data and the binary transformation can be understood at the same time. Only a few observations are needed to show the structure of the table. After that, the remaining observations can be summarized with the required statistical information.

#	Group A count	Group A binary	Group B count	Group B binary
1	$\text{count}_{A,1}$	0 or 1	$\text{count}_{B,1}$	0 or 1
2	$\text{count}_{A,2}$	0 or 1	$\text{count}_{B,2}$	0 or 1
3	$\text{count}_{A,3}$	0 or 1	$\text{count}_{B,3}$	0 or 1
⋮	⋮	⋮	⋮	⋮

Tasks

These are the requirements in order to pass each criterion. In order for a criterion to be passed, the corresponding task must be completed correctly. Some minor notation mistakes are allowed, as long as they do not change the procedure, the statistical information, the hypotheses, the calculations, the graph, the decision, or the interpretation.

C1

Organize the statistical information for both groups and both settings.

a) In the mean-count and proportion settings, define the population parameters for Group A and Group B.

b) Show a few sample rows using the required data-table format.

C2

Write the null and alternative hypotheses for all setup cases.

a) Mean-count and proportion settings using Group A as the benchmark and Group B as the test group.

b) Mean-count and proportion settings using Group B as the benchmark and Group A as the test group.

Use μ for the mean-count setting and p for the proportion setting.

C3

Classify each setup case as right-tailed, left-tailed, or two-tailed.

a) Draw an approximate rejection-region graph for each setup case, each graph must match the selected significance level and the correct tail direction.

Benchmark direction for C4–C7

After completing C1–C3, choose only one benchmark direction.

Selected benchmark direction: ______________

All work from C4 to C7 must use only this selected direction. Do not complete C4–C7 for the reverse direction.

C4

Apply the hypothesis test in the proportion setting for your selected benchmark direction.

a) Compute the test statistic.

b) Compare the test statistic with the appropriate critical value and state the decision about H_0.

c) Interpret the decision in the context given.

C5

Apply the hypothesis test in the mean-count setting for your selected benchmark direction.

a) Compute the test statistic.

b) Compare the test statistic with the appropriate critical value and state the decision about H_0.

c) Interpret the decision in the context given.

C7

Determine the minimum sample size needed to reject H_0 for the selected benchmark direction.

a) Find the minimum sample size for the proportion and the mean-count settings.

b) Verify that the sample size satisfies the rejection condition.

C6

Revisit the initial-sample-size decision and the minimum-sample-size decision for both settings.

a) Analyze the proportion and the mean-count settings with the initial sample size and the minimum sample size found in C7. For each case, state the decision, identify the possible error type, and explain its practical consequence in the context given.